INCOSE Social Systems Engineering WG · Integration Proposal · May 2026

Social Systems Engineering

Cross-disciplinary integration across INCOSE and organizational SE practice

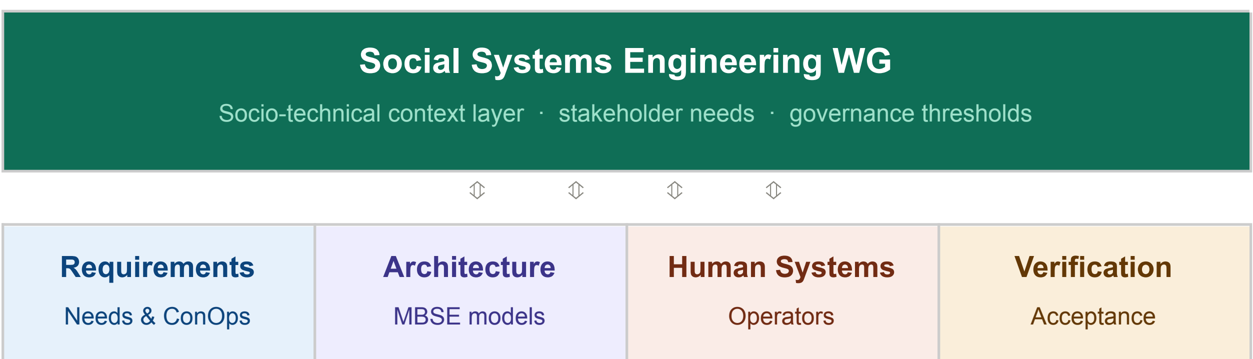
> The SSE WG's highest-value contribution is serving as the socio-technical context layer that activates and calibrates requirements and governance across all engineering disciplines. Every system that interacts with humans is a socio-technical system. SSE's integration mandate follows directly from this fact.

This proposal identifies four structural integration points: stakeholder context capture, socio-technical norms as requirements inputs, design and validation governance, and continuous run-time review. Each is mapped to INCOSE WG activity and to the SE project lifecycle.

1 · SSE in the INCOSE ecosystem

SSE is positioned as a cross-cutting context layer with bidirectional relationships to four core INCOSE working group clusters. The diagram reflects mutual dependency: SSE provides the socio-technical framing that makes technical outputs trustworthy; technical WGs provide the architectural vocabulary that gives SSE precision.

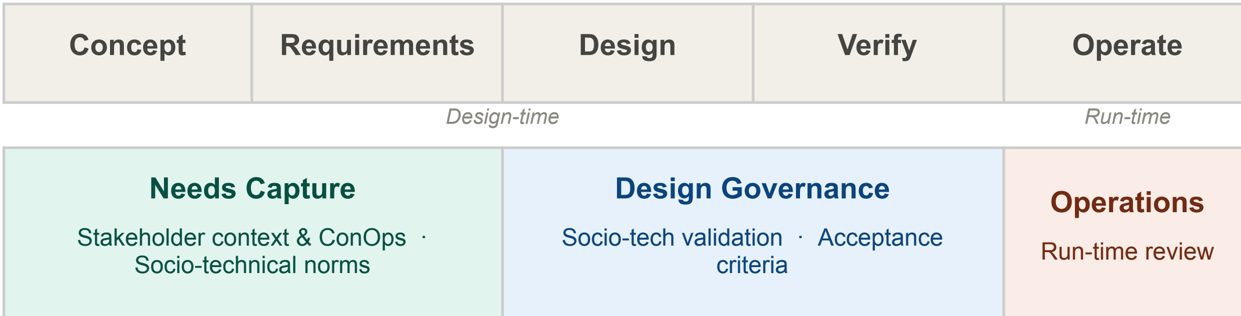
Figure 1 · INCOSE WG integration model



2 · Integration across the SE project lifecycle

Three SSE activity zones span the full SE lifecycle. The design-time zone covers needs capture, requirements input, and socio-technical validation. The run-time zone covers continuous review of whether deployed systems remain aligned with stakeholder intent. This division reflects a fundamentally different governance relationship with the system before and after deployment.

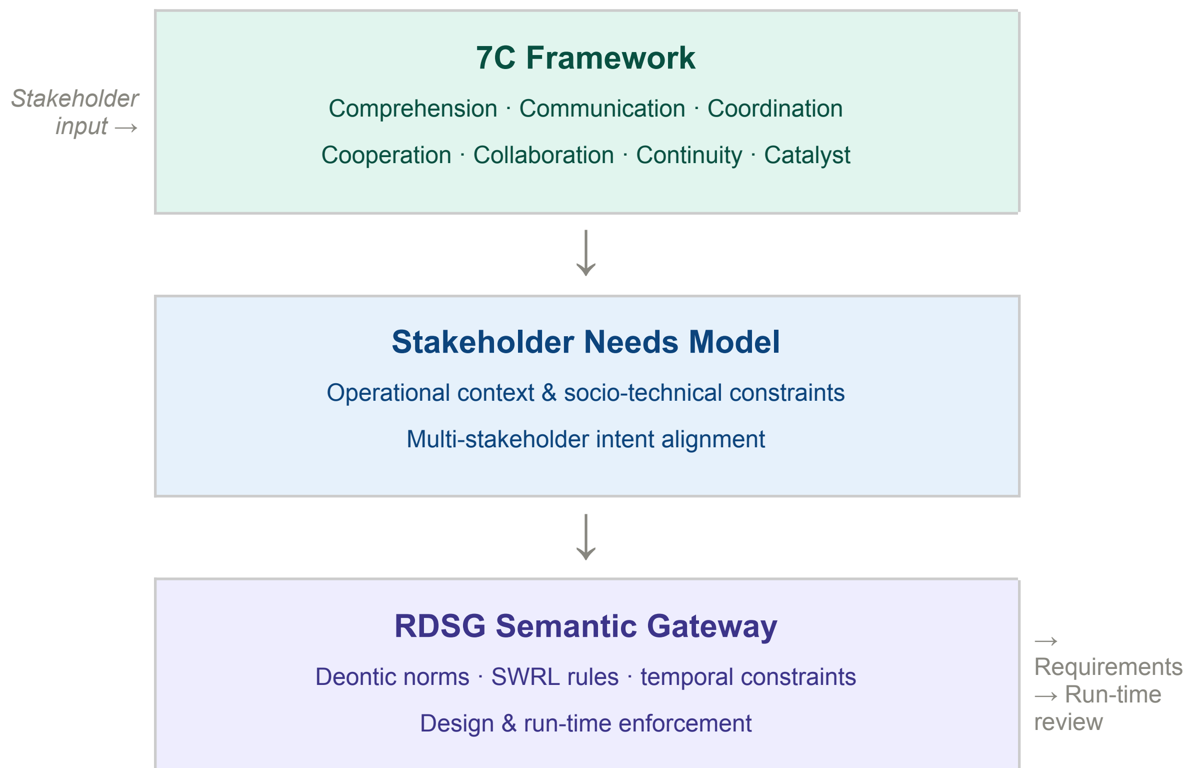
Figure 2 · SE lifecycle integration touchpoints



3 · The 7C framework as a requirements pipeline

The seven C-activities are not communication principles alone — they are a structured pipeline that produces artifacts feeding requirements governance. Comprehension and Communication generate operational context; Coordination and Cooperation surface competing stakeholder objectives; Collaboration, Continuity, and Catalyst define what must persist and adapt. Together, they produce a Stakeholder Needs Model that feeds the RDSG Semantic Gateway (a separate initiative & solution), where constraints are encoded as deontic norms enforced at design and run-time.

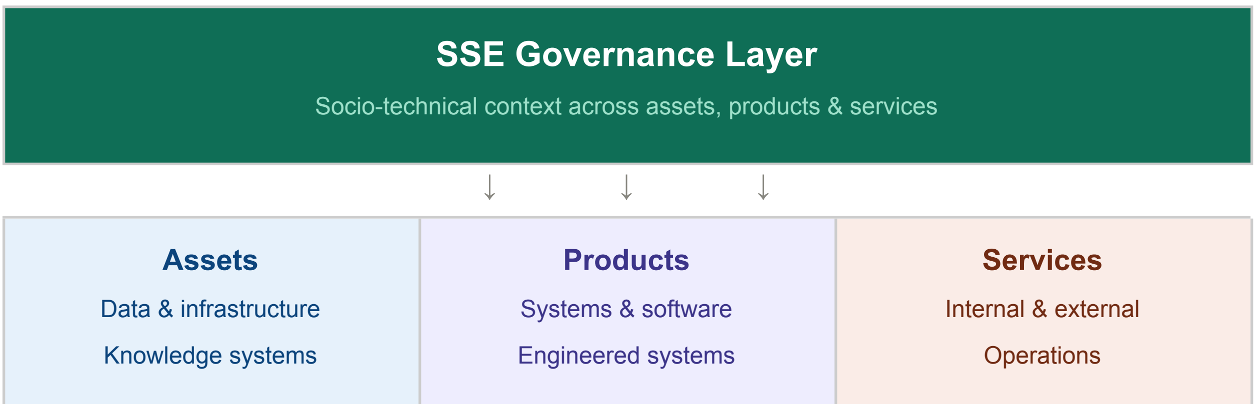
Figure 3 · 7C framework to semantic governance pipeline



4 · Organizational scope of SSE governance

In organizations that rely on systems engineering, SSE governance applies across the full portfolio of assets, products, and services — internal or external, involving one engineering discipline or many. The governance layer does not command these systems; it provides the continuous socio-technical context that keeps their designs aligned with the communities they serve across the full lifecycle.

Figure 4 · Organizational scope of SSE governance



5 · Recommended integration actions

Five actions represent the minimum viable path to establishing SSE as a structural participant in INCOSE's engineering practice:

1. **Cross-WG liaison roles**

   Embed SSE representatives in Requirements, Human Systems Integration, and Architecture WGs. This is the single highest-leverage structural change and requires no new WG formation or dedicated funding.

2. **Map 7C to ISO 15288 lifecycle stages**

   Demonstrate concretely where each C-element activates and what artifacts it produces or consumes. This creates a shared integration grammar between SSE and technical WGs.

3. **Define socio-technical threshold indicators**

   In collaboration with each technical WG, define what socio-technical drift looks like in their domain and at what level it triggers a governance review. These become SSE's run-time contribution.

4. **Pilot a joint STPA/UCA workshop**

   Run a single session with one technical WG demonstrating how SSE methods surface unsafe control action (UCA) candidates that purely technical STPA analysis misses. This is the strongest demonstration of additive value.

5. **RDSG v2.0 as reference implementation**
   The Requirements-Driven Semantic Gateway Architecture has SSE integration points structurally defined. Use it as a concrete demonstration vehicle for Figure 3 and as a pilot for broader cross-WG adoption.
